

LAB WORKBOOK

Assignment 1: The Multi-Locator Challenge

Tier 1: Core Fundamentals & Locators

1. Module Name and Experiment Title

Module: Tier 1 – Core Fundamentals & Locators

Experiment: The Multi-Locator Challenge

2. Problem Statement

Automate the login process of a web application using Selenium WebDriver. The username field must be located using By.ID, the password field using By.NAME, and the login button using By.XPATH. After successful login, verify that the resulting page URL contains '/inventory.html'.

3. Objective

To practice identifying and interacting with web elements using different Selenium locator strategies and to validate successful navigation after a login operation.

4. Tools, Software and Concepts Used

Category	Details
Programming Language	Python 3
Automation Tool	Selenium WebDriver
Browser	Google Chrome
IDE/Editor	VS Code / PyCharm
Website	SauceDemo (https://www.saucedemo.com/)
Concepts	WebDriver, element locators, By.ID, By.NAME, By.XPATH, assertions, URL validation

5. Source Code / Implementation

Install Selenium if required using: `pip install selenium`. Then run the following program.

```
from selenium import webdriver
from selenium.webdriver.common.by import By

# Launch Chrome
driver = webdriver.Chrome()

try:
    # Open SauceDemo login page
    driver.get("https://www.saucedemo.com/")
```

```

# Locate username using By.ID
username = driver.find_element(By.ID, "user-name")
username.send_keys("standard_user")

# Locate password using By.NAME
password = driver.find_element(By.NAME, "password")
password.send_keys("secret_sauce")

# Locate login button using By.XPATH
login_button = driver.find_element(By.XPATH, "//input[@id='login-button']")
login_button.click()

# Validate the resulting URL
assert "/inventory.html" in driver.current_url, (
    "Login failed or inventory page was not opened."
)

print("Login successful.")
print("Validation passed: /inventory.html is present in the URL.")

finally:
    driver.quit()

```

6. Implementation Steps

1. Open the SauceDemo login page in Chrome using Selenium WebDriver.
2. Find the username field with By.ID and enter the username standard_user.
3. Find the password field with By.NAME and enter the password secret_sauce.
4. Find the Login button with By.XPATH and click it.
5. Read the current browser URL using driver.current_url.
6. Assert that the URL contains /inventory.html.
7. Close the browser after execution.

7. Output

After successful execution, the browser should navigate to the SauceDemo inventory page and the console should display successful validation.

Figure 1: Successful SauceDemo login and URL validation output.

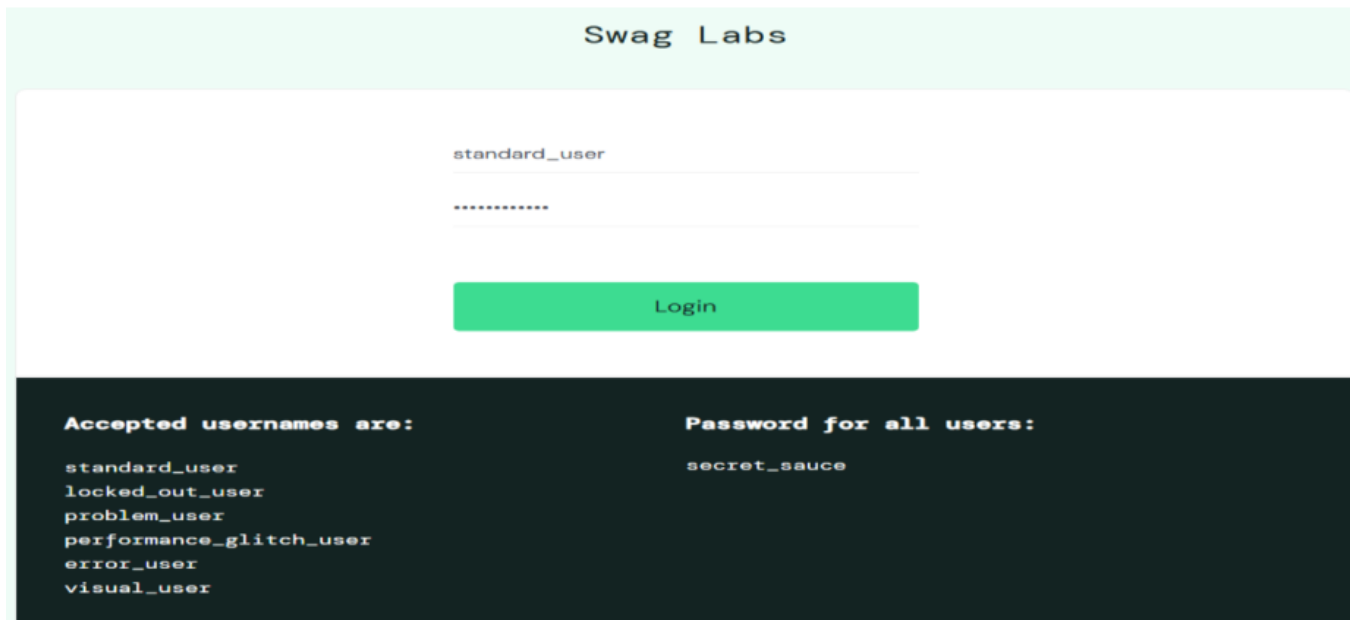
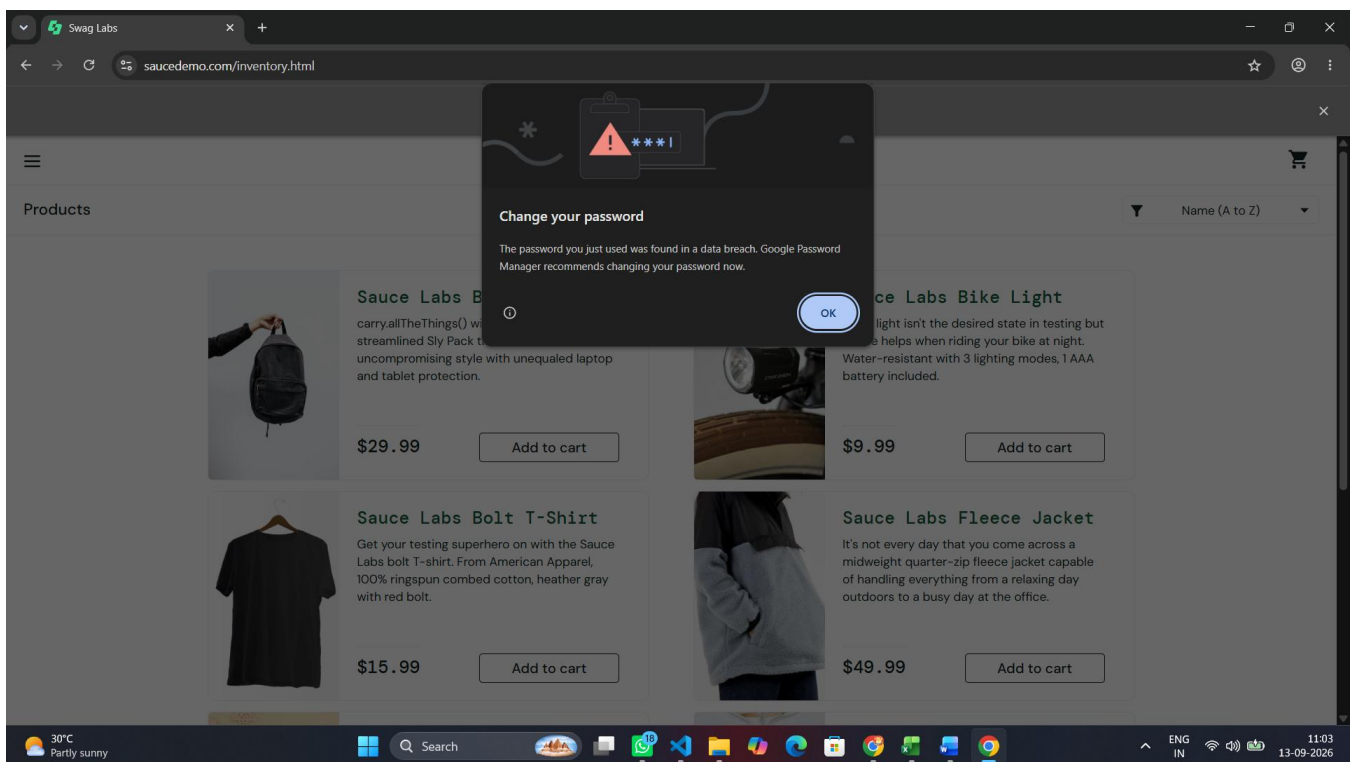


Figure 2: Products inventory page



8. Result

The SauceDemo login process was successfully automated using three different Selenium locator strategies. The resulting URL was verified to contain /inventory.html.

9. Observation

Selenium provides multiple locator strategies for identifying web elements. Different locators can be selected according to the attributes available in the HTML of the page.

10. Conclusion

This experiment demonstrated the use of By.ID, By.NAME, and By.XPATH for locating web elements and showed how an assertion can be used to validate successful browser navigation.